

179. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8

179.1 Overview

This clause specifies the 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 PMDs and the associated baseband media. The PMDs provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links on one, two, four, or eight lanes, respectively, over twinaxial copper cable. The specifications for the four PMDs are similar, except for the number of lanes and associated parameters, and the MDI.

There are four associated annexes. Annex 179A provides information on parameters that might not be testable in an implemented system, since the test points they are associated with are typically inaccessible. Annex 179B includes specifications and reference insertion loss for test fixtures. Annex 179C specifies MDIs. Annex 179D describes host and cable assembly types.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate sublayers (as specified in Table 179–1 through Table 179–4) and to the medium through the appropriate MDI, as illustrated in Figure 179–1.

PMDs defined in this clause conform to at least one of three host classes: HH, HN and HL. The PMD electrical specifications in 179.9.4 and 179.9.5 are different for each host class. Annex 179A provides information that can be used for designing host channels that enable meeting the electrical specifications for each host class.

Cable assemblies defined in this clause conform to one of four classes that differ by the maximum insertion loss, as specified in 179.11.

Operation of Ethernet links is provided for specific combinations of host classes and cable assembly classes, as listed in Table 179–21 and Table 179A–4.

Table 179–1—Physical Layer clauses associated with the 200GBASE-CR1 PMD

Associated clause	200GBASE-CR1
73—AN	Required
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120F—200GAUI-2 C2C	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.5.1.

^c When implemented in a 200GBASE-CR1 PHY, the signaling rate range is constrained as specified to 120.1.4.

^d A 200GBASE-CR1 PHY may include one instance of 200GAUI-n C2C as described in 176B.5.1.

Table 179–2—Physical Layer clauses associated with the 400GBASE-CR2 PMD

Associated clause	400GBASE-CR2
73—AN	Required
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—400GBASE-R PCS	Optional
120—400GBASE-R BM-PMA	Conditional ^b

Table 179–2—Physical Layer clauses associated with the 400GBASE-CR2 PMD (*continued*)

Associated clause	400GBASE-CR2
120B—400GAUI-16 C2C ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.6.1.

^c When used within a 400GBASE-CR2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d A 400GBASE-CR2 PHY may include one instance of 400GAUI-n C2C as described in 176B.6.1.

Table 179–3—Physical Layer clauses associated with the 800GBASE-CR4 PMD

Associated clause	800GBASE-CR4
73—AN	Required
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a

^a A 800GBASE-CR4 PHY may include one instance of 800GAUI-n C2C as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Table 179–4—Physical Layer clauses associated with the 1.6TBASE-CR8 PMD

Associated clause	1.6TBASE-CR8
73—AN	Required
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
170—1.6 Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII extender	Optional
175—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a

^a A 1.6TBASE-CR8 PHY may include one instance of 1.6TAUI-n C2C as described in 176B.8.1.

^b The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^c If a 1.6TAUI-n is implemented in a PHY, additional 1.6TBASE-R SM-PMA sublayers are required according to 176B.8.1.

Figure 179–1 shows the relationship of the PMD and MDI, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model.

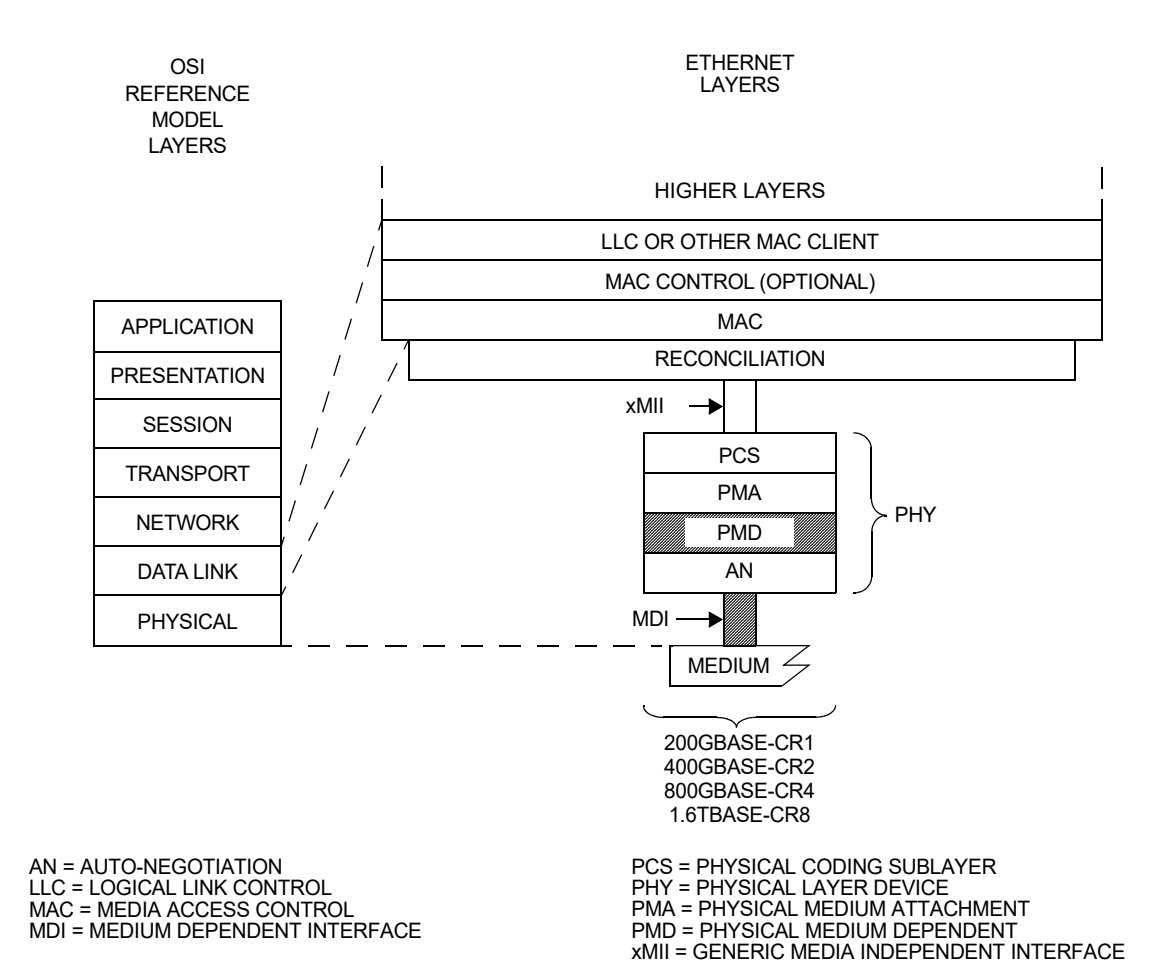


Figure 179-1—200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

179.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.6.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the PMA adjacent to the PMD using the method described in 174A.9, with BER_{added} equal to 1.6×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the BLER of 1.45×10^{-11} (see 174A.6), measured at the PCS using the method described in 174A.11, with BER_{added} equal to 8×10^{-6} .

179.3 Conventions

This clause describes four PMDs, 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8, which have one, two, four, and eight lanes, respectively. For efficient description, the

parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n=1$ for 200GBASE-CR1, $n=2$ for 400GBASE-CR2, $n=4$ for 800GBASE-CR4, and $n=8$ for 1.6TBASE-CR8.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n-1$.

Within this clause, the unqualified term “PMD” refers to any of 200GBASE-CR1 PMD, 400GBASE-CR2 PMD, 800GBASE-CR4 PMD, or 1.6TBASE-CR8 PMD.

Within this clause, the term FEC refers to the Reed-Solomon code within the Clause 119 PCS for 200GBASE-CR1 and 400GBASE-CR2, the Clause 172 PCS for 800GBASE-CR4, and the Clause 175 PCS for 1.6TBASE-CR8.

179.4 Service interfaces

This subclause specifies the services provided by the PMD. The PMD service interface is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (PMA immediately above the PMD)). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-CR1 and 400GBASE-CR2, in 169.3 for 800GBASE-CR4, and in 174.3 for 1.6TBASE-CR8. The nominal signaling rate for each symbol stream in `PMD:IS_UNITDATA_i.request` and `PMD:IS_UNITDATA_i.indication`, where $i = 0$ to $n-1$, is 106.25 GBd.

The `SIGNAL_OK` parameter of the `PMD:IS_SIGNAL.indication` primitive (see 116.3.3.3) corresponds to the variable `rts_status` of the ready-to-send (RTS) function, as defined in 178B.8.2.1. When `SIGNAL_OK` is either `IN_PROGRESS` or `FAIL`, the `rx_symbol` parameters of `PMD:IS_UNITDATA_i.indication` on all lanes are unspecified.

The PMD receives signal status information from the PMD client using the `PMD:IS_SIGNAL.request` primitive (see 116.3.3.4).

179.5 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive `AN_LINK.indication` defined in 73.9 (See 119.6, 172.6, and 175.7).

200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4 and 1.6TBASE-CR8 PHYs may be extended using 200GAUI-n C2C, 400GAUI-n C2C, 800GAUI-n C2C, and 1.6TAUI-n C2C, respectively, as physical instantiations of the inter-sublayer service interface between devices. If such extension is used, then the `AN_LINK(link_status).indication` is relayed from the device with the PCS sublayer to the device with the AN sublayer by means at the discretion of the implementer. As examples, the implementer may employ system management or use a dedicated electrical signal to relay the state of `link_status` as indicated by the PCS sublayer on one device to the AN sublayer on the other device.

179.6 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN sublayers including the medium in one direction shall be no more than the maximum delays listed in Table 179–5. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 179–5—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GBASE-CR1	14 848	29	74.24
400GBASE-CR2	29 696	58	74.24
800GBASE-CR4	59 392	116	74.24
1.6TBASE-CR8	118 784	232	74.24

^a One bit time is equal to 5 ps for 200GBASE-CR1, 2.5 ps for 400GBASE-CR2, 1.25 ps for 800GBASE-CR4, and 625 fs for 1.6TBASE-CR8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GBASE-CR1, 1.28 ns for 400GBASE-CR2, 640 ps for 800GBASE-CR4, and 320 ps for 1.6TBASE-CR8. See 1.4.459 for the definition of pause_quanta.

Descriptions of overall system delay constraints can be found in 116.4 for 200GBASE-CR1 and 400GBASE-CR2, in 169.4 for 800GBASE-CR4, and in 174.4 for 1.6TBASE-CR8.

179.7 Skew constraints

The Skew (relative delay) between the PMD lanes is kept within limits so that the information on the PCS lanes can be reassembled by the PCS. The Skew Variation is also limited to ensure that a given PCS lane always traverses the same PMD lane.

179.7.1 Skew Constraints for 200GBASE-CR1 and 400GBASE-CR2

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5. Skew Variation at SP2 to SP5 is specified only for 400GBASE-CR2. For 200GBASE-CR1, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps. The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

179.7.2 Skew Constraints for 800GBASE-CR4 and 1.6TBASE-CR8

Skew and Skew Variation are defined in 169.5 for 800GBASE-CR4 and in 174.5 for 1.6TBASE-CR8. They are specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5 for 800GBASE-CR4, and in Figure 174–5 and Figure 174–6 for 1.6TBASE-CR8.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5 and 174.5.

179.8 PMD functional specifications

179.8.1 Reference test points

For purposes of system conformance, the PMD sublayer is standardized at the test points described in Table 179–6.

The test points are illustrated at their corresponding link locations in Figure 179–2, which shows one direction of a 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8 link. Each differential pair in the cable is separately shielded. The signal shields are connected to ground contacts in the MDI plug connectors on both ends of the cable assembly, creating a common ground reference for both PMDs.

Table 179–6—Test points

Test point	Description
TP0d	Represents the transmitter output at the device-to-package interface. This point is not intended to be accessible for measurements.
TP1	The input of a cable assembly test fixture (see 179B.3), corresponding to MDI signals $SL_i<p>$ and $SL_i<n>$, used in cable assembly specifications.
TP2	The output of a TP2 or TP3 test fixture (see 179B.2), corresponding to MDI signals $SL_i<p>$ and $SL_i<n>$. Host output measurements and tests defined in 179.9.4 are made at this point.
TP3	The input of a TP2 or TP3 test fixture (see 179B.2), corresponding to MDI signals $DL_i<p>$ and $DL_i<n>$. Host input measurements and tests defined in 179.9.5 are made at this point.
TP4	The output of a cable assembly test fixture (see 179B.3), corresponding to MDI signals $DL_i<p>$ and $DL_i<n>$, used in cable assembly specifications.
TP5d	Represents the receiver input at the device-to-package interface. This point is not intended to be accessible for measurements.
NOTE 1—The channel between TP0d and TP5d includes the transmitter and receiver packages and PCBs and the cable assembly.	
NOTE 2—Cable assembly measurements for the specifications in 179.11 are made between TP1 and TP4 as illustrated in Figure 179A–1. These measurements include a mated connector pair at each end of the cable assembly.	
NOTE 3—A mated connector pair is accounted for in transmitter specifications at TP2 and in receiver specifications at TP3. The recommended maximum insertion loss from TP0d to TP2 or from TP3 to TP5d, excluding the test fixtures, is provided in 179A.4.	
NOTE 4—Test fixture connector labeling may differ from the MDI signal names.	

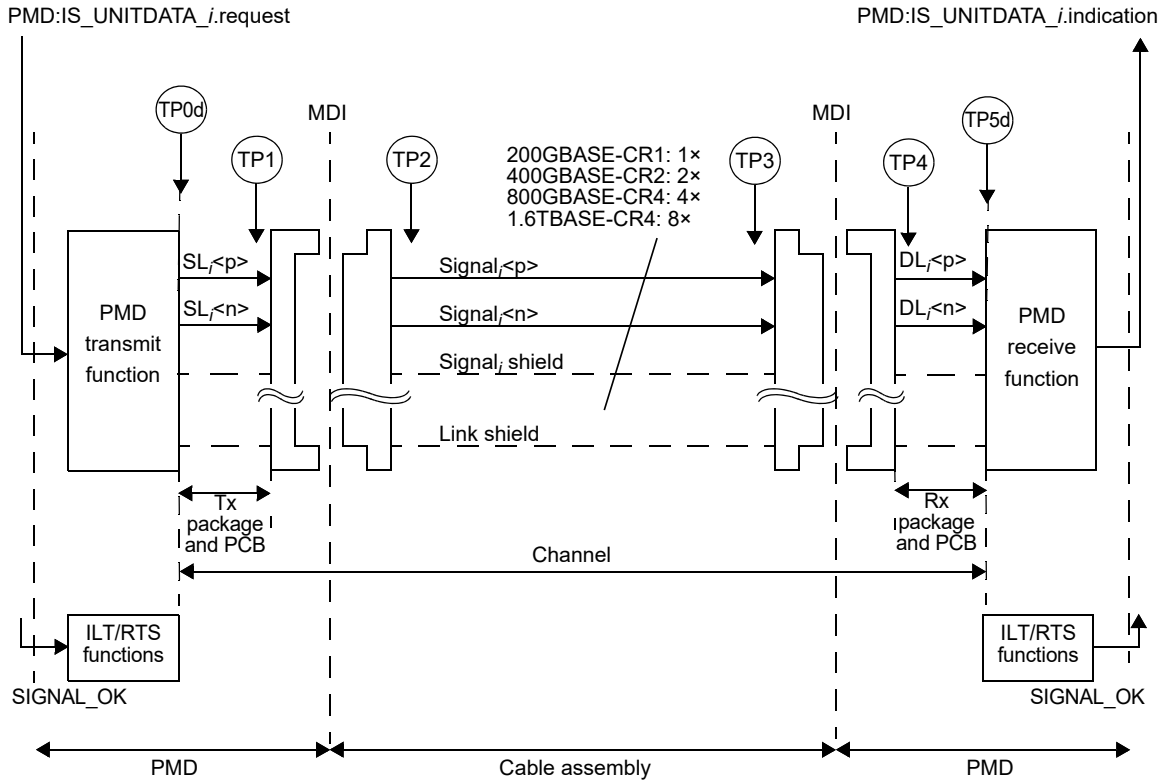


Figure 179-2—200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4 or 1.6TBASE-CR8 link diagram

Annex 179A provides information on parameters associated with test points TP0d and TP5d, which are typically inaccessible in an implemented system.

179.8.2 PMD transmit function

The PMD transmit function has two operating modes: DATA and TRAINING. The operating mode is controlled by the `tx_mode` variable of the inter-sublayer link training (ILT) function (see 179.8.9): it is DATA when `tx_mode` = DATA, and TRAINING otherwise.

When operating in DATA mode, the PMD transmit function shall convert the `tx_symbol` parameters of the PMD service interface message `PMD:IS_UNITDATA_i.request` on each lane into a stream of PAM4 symbols for transmission as electrical signals on the corresponding contacts of the MDI (see 179.12). The PAM4 symbol values 0, 1, 2, and 3 correspond to the `tx_symbol` values zero, one, two, and three, respectively. When operating in TRAINING mode, the PAM4 symbol stream on each lane is provided by the ILT function (see 178B.7).

Each lane has two output signals, $SL_i<p>$ and $SL_i<n>$, which create a differential signal $SL_i<p> - SL_i<n>$ and a common-mode signal $(SL_i<p> + SL_i<n>) / 2$. The differential and common-mode signals on each lane comply with the transmit electrical specifications in 179.9.4.

The polarity of the PMD output on each of the lanes is either normal, where the highest differential output voltage corresponds to the PAM4 symbol 3 and the lowest differential output voltage corresponds to the PAM4 symbol 0, or inverted, where the highest differential output voltage corresponds to the PAM4 symbol 0 and the lowest differential output voltage corresponds to the PAM4 symbol 3. If training is enabled, either normal or inverted output polarity may be used, since the PMD receiver in the peer interface detects the polarity and corrects it if necessary (see 179.8.3). If training is disabled, the output polarity of each lane shall be normal. Output polarity may be controlled by management in an implementation dependent manner.

179.8.3 PMD receive function

The PMD receive function converts the differential electrical signal on each of the lanes on the MDI ($DL_{i<p>} - DL_{i<n>}$) into a symbol stream for delivery to the PMD service interface using the messages `PMD:IS_UNITDATA_i.indication(rx_symbol)`.

The polarity of the PMD input on each of the lanes is either normal, where the highest differential input voltage at the MDI input corresponds to `rx_symbol = three` and the lowest differential input voltage corresponds to `rx_symbol = zero`, or inverted, where the highest differential input voltage at the MDI corresponds to `rx_symbol = zero` and the lowest differential input voltage corresponds to `rx_symbol = three`. If training is enabled, the PMD shall detect the polarity during training and correct it if necessary (see 178B.7.7). If training is disabled, the input polarity of each lane shall be normal. Input polarity may be controlled by management in an implementation dependent manner.

The PMD receive function also recovers an appropriate clock from the input signal on each lane for timing of the delivery of `PMD:IS_UNITDATA_i.indication` messages to the PMD client.

179.8.4 PMD global signal detect function

The PMD global signal detect function sets the variable `Global_PMD_signal_detect` to the value of `remote_rts` in the RTS function (see 178B.8.2.1). When `mr_training_enable` is true, this variable indicates the status of the RTS function in the peer interface, as communicated through the training protocol (see 179.8.9). When `mr_training_enable` is false this variable is always true.

179.8.5 PMD lane-by-lane signal detect function

The variable `PMD_signal_detect_i` is the logical AND of the variables `local_rx_ready` and `remote_rx_ready` in lane *i* of the ILT function. These variables are set independently on each lane.

179.8.6 PMD global transmit disable function

The PMD global transmit disable function is optional. If this function is supported, it shall meet the following requirements.

Setting the `Global_PMD_transmit_disable` variable to 1 has the same effect as if `tx_disable` (see 179.8.7) is set to 1 on all lanes ($i = 0$ to $n-1$). `Global_PMD_transmit_disable` does not overwrite the `tx_disable` or `PMD_transmit_disable_i` variables.

179.8.7 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function enables selective disabling of the transmitters in each lane. This function is used by the ILT function. It may optionally be provided through control variables `PMD_transmit_disable_i`.

If either `PMD_transmit_disablei` or the variable `tx_disable` in lane *i* of the ILT function (see 178B.8.2.1) are set to one, the transmitter output on lane *i* of the PMD shall not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 179–7.

179.8.8 PMD fault functions

PMD transmit fault indication is optional. The faults detected by this function are implementation specific. A fault is indicated by setting the variable `PMD_transmit_fault` to one. The assertion of `Global_PMD_transmit_disable` is not considered a transmit fault. However, if `PMD_transmit_fault` is set to one, then `Global_PMD_transmit_disable` should also be set to one.

PMD receive fault indication is optional. The faults detected by this function are implementation specific. A fault is indicated by setting the variable `PMD_receive_fault` to one.

The variable `PMD_fault` is the logical-OR of `PMD_receive_fault`, `PMD_transmit_fault`, and any other implementation-specific fault indications.

179.8.9 Autonomous path startup (APSU) functions

The PMD shall provide the RTS function and the ILT function with training frame format E1, as specified in Annex 178B.

When the variable `mr_training_enable` is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), control the PMD transmitter output on each lane based on requests from the peer, indicate the receiver state, and coordinate the transition of the PMD transmit function to DATA mode. The PMD transmit function is configured in response to the requests in the incoming training frames as specified in 179.9.4.2.3 through 179.9.4.2.5. The PMD transmit function supports the coefficient indexes `k_list` = {–3, –2 –1, 0, 1} and the initial conditions PRESET 1 through PRESET 6 and INITIALIZE (see Table 179–8).

When `tx_mode` = `LOCAL_PATTERN` (see 178B.8.3.1), the PMD transmits PRBS31Q (see 176.7.4.2).

The default value of `max_wait_timer_duration` is 30, corresponding to a duration of 30 seconds for `max_wait_timer` (see 178B.8.3.3).

179.8.10 PMD reset function

If the variable `PMD_reset` is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

PMD reset shall reset the ILT and RTS functions associated with the PMD (see 179.8.9).

179.9 PMD electrical characteristics

PMD compliance with a host class (HH, HN, or HL) is defined by meeting the requirements of 179.9.4 and 179.9.5 for that host class on all MDI lanes (see 179.9.2).

NOTE—A PMD can comply with one or more host classes.

179.9.1 AC-coupling

Interoperability between PMD components operating from different supply voltages is facilitated by AC-coupling within the cable assembly as specified in 179.11.

179.9.2 Signal paths

The MDI transmit and receive paths are point-to-point connections. Each path corresponds to one MDI lane and is composed of two complementary signals, which form a balanced differential pair.

For 200GBASE-CR1, there is one differential lane in each direction for a total of two pairs, or four connections. For 400GBASE-CR2, there are two differential lanes in each direction for a total of four pairs, or eight connections. For 800GBASE-CR4, there are four differential lanes in each direction for a total of eight pairs, or 16 connections. For 1.6TBASE-CR8, there are eight differential lanes in each direction for a total of 16 pairs, or 32 connections.

179.9.3 Reference impedance

The reference impedance for differential specifications is $92.5\ \Omega$. The reference impedance for common-mode specifications is $23.125\ \Omega$.

NOTE—Calculation of modal S-parameters may require transformation of the measured single-ended S-parameters to the specified reference impedance. See 178A.1.3.

Unless specified otherwise, specified parameters are measured on a load or termination equal to the reference impedance.

179.9.4 Transmitter characteristics

The transmitter on each lane shall meet the specifications at TP2 given in Table 179–7 and detailed in the referenced subclauses.

Unless specified otherwise, transmitter signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with AC-coupled connection from TP2 to $50\ \Omega$ single-ended loads in the test equipment.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP2 where each conductor of the differential pair is AC-coupled to a $50\ \Omega$ termination.

Recommended transmitter characteristics at TP0d are provided in 179A.2.

Table 179–7—Summary of transmitter specifications at TP2

Parameter	Reference	Value	Units
Signaling rate (range)	179.9.4.1	106.25 ± 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.2	0.03	V
Transmitter disabled		1	V
Transmitter enabled			
DC common-mode voltage (range)	176D.8.2	0.2 to 1	V
AC common-mode peak-to-peak voltage (max)	176D.8.2	0.032	V
Low-frequency, $V_{CM_{LF}}$		0.08	V
Full-band, $V_{CM_{FB}}$			
Effective return loss, ERL (min)	179.9.4.8.1	7.3	dB
Common-mode to common-mode return loss	179.9.4.8.3	Table 179–11	—
Common-mode to differential-mode return loss	179.9.4.8.3	Table 179–11	—

Table 179–7—Summary of transmitter specifications at TP2 (*continued*)

Parameter	Reference	Value	Units
Transmitter steady-state voltage, v_f (min)	179.9.4.2.2		
Host class HL		0.389	V
Host class HN		0.385	V
Host class HH		0.379	V
Transmitter steady-state voltage, v_f (max)	179.9.4.2.2	0.5	V
Linear fit pulse peak ratio, R_{peak} (min)	179.9.4.2.2		
Host class HL		0.44	—
Host class HN		0.332	—
Host class HH		0.269	—
Level separation mismatch ratio, R_{LM} (min)	179.9.4.3	0.95	—
Transmitter output equalization			
absolute value of step size for all taps (range)	179.9.4.2.4	0.005 to 0.025	—
value at minimum state for $c(-3)$ (max)	179.9.4.2.5	−0.06	—
value at maximum state for $c(-2)$ (min)	179.9.4.2.5	0.12	—
value at minimum state for $c(-1)$ (max)	179.9.4.2.5	−0.34	—
value at minimum state for $c(0)$ (max)	179.9.4.2.5	0.5	—
value at minimum state for $c(1)$ (max)	179.9.4.2.5	−0.2	—
Signal-to-noise-and-distortion ratio, SNDR (min)	179.9.4.6	Table 179–9	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	179.9.4.4	26.7	dB
Output jitter (max)	179.9.4.7		
JH_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
JH_{4u}		0.118	UI

179.9.4.1 Transmitter signaling rate

The transmitter signaling rate shall be within the range specified in Table 179–7.

NOTE—If the Physical Layer includes an AUI-C2C interface, the signaling rate is derived from the signaling rate of that interface.

179.9.4.2 Transmitter output equalization

The PMD transmit function includes programmable equalization to compensate for the frequency-dependent loss of the channel and facilitate data recovery at the receiver. The functional model for the transmit equalizer on each lane is the five-tap transversal filter shown in Figure 179–3.

The state of the transmit equalizer may be manipulated via the ILT function (see 179.8.9) or via the PMD management interface. The transmit function responds to a set of commands issued by the peer interface receive function and conveyed by a back-channel communications path. This command set includes instructions to increment, decrement, hold, or set to zero a selected coefficient $c(k)$, where $k = -3$ to 1. In addition, it includes commands to set all coefficients to one of six initial conditions.

In response, the transmit function relays status information to the peer interface receive function. The status messages indicate the selected coefficient index k , the status of the coefficient $c(k)$ (UPDATED, NOT UPDATED, or COEFFICIENT AT LIMIT), and the status of the initial condition (UPDATED or NOT UPDATED).

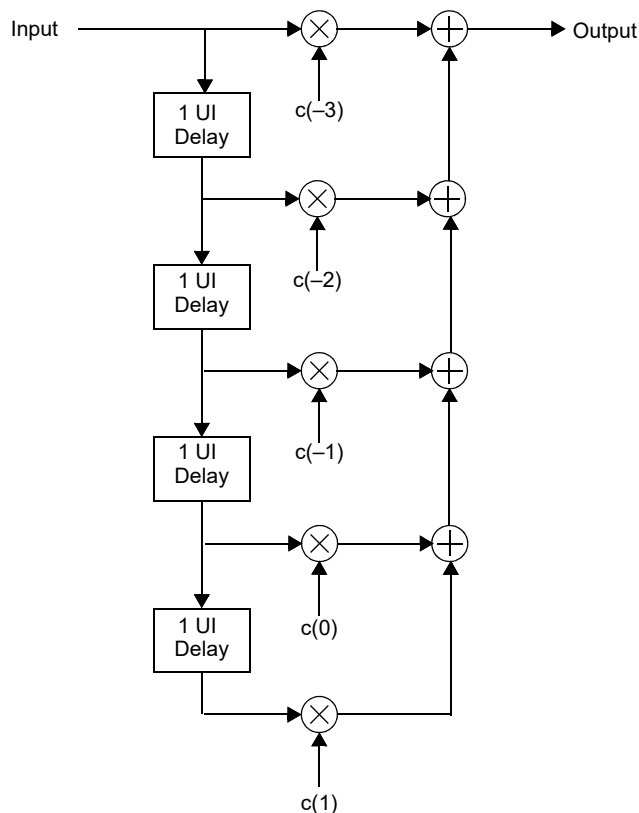


Figure 179-3—Transmit equalizer functional model

179.9.4.2.1 Linear fit to the measured waveform

The following procedure is used to determine the linear fit pulse response, linear fit error, and normalized transmitter coefficient values for a specific configuration of the transmit equalizer.

Set the transmitter under test to transmit the PRBS13Q test pattern (defined in 120.5.11.2.1). Capture at least one complete cycle of the test pattern at TP2, as specified in 85.8.3.3.4. The reference clock recovery unit (CRU) used in the measurement acts as a high-pass jitter filter with a corner frequency of 4 MHz and a slope of 20 dB/decade.

In the following calculations, M should be an integer greater than or equal to 32. Interpolation of the captured waveform may be used to achieve this.

Compute the linear fit pulse response $p(k)$ and linear fit error $e(k)$, $k=1$ to $M \times N_p$, from the captured waveform, as specified in 85.8.3.3.5, with $N_p = 700$ and $D_p = 4$, where the aligned symbols $x(n)$ are assigned normalized amplitudes -1 , $-ES$, ES , and 1 to represent the PAM4 symbol values 0, 1, 2, and 3 respectively. ES is defined as $(|ES1| + |ES2|)/2$ where $ES1$ and $ES2$ are calculated according to 120D.3.1.2.

Define $r(k)$ to be the linear fit pulse response $p(k)$ when the transmit equalizer coefficients have been set to the PRESET 1 values.

NOTE— $r(k)$ is required for calculation of the normalized transmit equalizer coefficients from a measurement. The same $r(k)$ is used in this calculation for any configuration of the transmit equalizer, but it needs to be calculated from a separate measurement.

For each value of m in the range $-M/2$ to $M/2 - 1$ when M is even and $-(M-1)/2$ to $(M-1)/2$ when M is odd:

- a) Define an $M \times N_p$ -by-5 matrix R_m . The elements of R_m are assigned values per Equation (179-1) where $i = -3$ to 1 and $j = 1$ to $M \times N_p$.

$$R_m(j, i+4) = \begin{cases} r(m+j-M \times i) & \text{if } 1 \leq m+j-M \times i \leq M \times N_p \\ 0 & \text{otherwise} \end{cases} \quad (179-1)$$

- b) The normalized coefficients of the transmit equalizer for phase m are computed using Equation (179-2).

$$\begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} = (R_m^T R_m)^{-1} R_m^T \begin{bmatrix} p(1) \\ \dots \\ p(M \times N_p) \end{bmatrix} \quad (179-2)$$

- c) The linear fit pulse response for phase m is reconstructed from the matrix R_m and the normalized coefficients using Equation (179-3).

$$\begin{bmatrix} p_m(1) \\ \dots \\ p_m(M \times N_p) \end{bmatrix} = R_m \begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} \quad (179-3)$$

- d) $\varepsilon^2(m)$ is computed using Equation (179-4).

$$\varepsilon^2(m) = \sum_{k=1}^{M \times N_p} (p(k) - p_m(k))^2 \quad (179-4)$$

The normalized transmit equalizer coefficients $c(i)$ for a given linear fit pulse $p(k)$ are the values $c_m(i)$ for the value of m that minimizes $\varepsilon^2(m)$.

179.9.4.2.2 Steady-state voltage and linear fit pulse peak ratio

The linear fit pulse peak v_{peak} and steady-state voltage v_f are defined using the linear fit pulse response $p(1)$ through $p(M \times N_p)$ with $N_p = 400$, measured with transmit equalizer set to PRESET 1 (no equalization). The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 179.9.4.2.1.

v_{peak} is defined as the maximum value of $p(k)$.

v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_p)$ divided by M .

Systems with transmitters having steady-state voltage higher than the maximum specified in Table 179–7 are considered engineered links. It is the system integrator’s responsibility to verify that the transmitter, receiver, and channel are compatible.

The linear fit pulse peak ratio R_{peak} is defined by Equation (179–5).

$$R_{peak} = \frac{v_{peak}}{v_f} \quad (179-5)$$

179.9.4.2.3 Coefficient initialization

When the coefficient update state diagram (Figure 178B–12) is in either the OUT_OF_SYNC state or the NEW_IC state, the coefficients of the transmit equalizer shall be configured to values within the ranges specified in Table 179–8 (according to the value of the variable `ic_req`, which is defined in 178B.8.3.1). These requirements apply upon the assertion of initial condition status of UPDATED.

Table 179–8—Coefficient initial conditions

ic_req	c(–3)	c(–2)	c(–1)	c(0)	c(1)
INITIALIZE	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.75 ± 0.025	0 ± 0.025
PRESET 1 ^a	0	0	0	1	0
PRESET 2	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.5 ± 0.025	0 ± 0.025
PRESET 3	0 ± 0.025	0 ± 0.025	–0.075 ± 0.025	0.75 ± 0.025	0 ± 0.025
PRESET 4	0 ± 0.025	0.05 ± 0.025	–0.2 ± 0.025	0.75 ± 0.025	0 ± 0.025
PRESET 5	–0.025 ± 0.025	0.075 ± 0.025	–0.25 ± 0.025	0.65 ± 0.025	0 ± 0.025
PRESET 6	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.75 ± 0.025	0 ± 0.025

^a PRESET 1 is the reference for the calculation of the normalized coefficients of the transmit equalizer (see 179.9.4.2.1). As a result, the normalized coefficients for PRESET 1 do not include any tolerances.

179.9.4.2.4 Coefficient step size

The change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to INCREMENT shall be between 0.005 and 0.025, and the change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to DECREMENT shall be between –0.025 and –0.005. It is recommended that the same step size is used for all coefficients.

The coefficients other than $c(\text{coef_sel})$ are not expected to change. The absolute change in any coefficient other than $c(\text{coef_sel})$ shall be less than 0.005.

The change in the normalized transmit equalizer coefficient is defined to be the difference in the value measured prior to the assertion of the INCREMENT or DECREMENT request (i.e., *coef_req* is HOLD) and the value upon the assertion of a coefficient status of UPDATED.

179.9.4.2.5 Coefficient range

When sufficient INCREMENT or DECREMENT requests have been received for a given coefficient, the coefficient reaches a lower or upper bound based on the range of that coefficient or the combination of coefficients.

The sum of the absolute values of $c(-3)$, $c(-2)$, and $c(-1)$, $c(0)$, and $c(1)$ shall be less than or equal to 1.

With $c(-3)$, $c(-2)$, and $c(-1)$ set to zero and both $c(0)$ and $c(1)$ having received sufficient DECREMENT requests so that they are at their respective minimum values, $c(1)$ shall be less than or equal to -0.2 .

With $c(-3)$, $c(-2)$, $c(-1)$, and $c(1)$ set to zero and having received sufficient DECREMENT requests so that it is at its minimum value, $c(0)$ shall be less than or equal to 0.5 .

With $c(-3)$, $c(-2)$, and $c(1)$ set to zero and both $c(-1)$ and $c(0)$ having received sufficient DECREMENT requests so that they are at their respective minimum values, $c(-1)$ shall be less than or equal to -0.34 .

With $c(-3)$, $c(-1)$, and $c(1)$ set to zero, $c(0)$ having received sufficient DECREMENT requests so that it is at its minimum value, and $c(-2)$ having received sufficient INCREMENT requests so that it is at its maximum value, $c(-2)$ shall be greater than or equal to 0.12 .

With $c(-2)$, $c(-1)$, and $c(1)$ set to zero and both $c(-3)$ and $c(0)$ having received sufficient DECREMENT requests so that they are at their respective minimum values, $c(-3)$ shall be less than or equal to -0.06 .

NOTE—Any of the coefficients $c(-3)$, $c(-2)$, $c(-1)$, or $c(1)$ can be set to zero by asserting a coefficient request of NO EQUALIZATION for that coefficient, using the control function specified in 179.8.9, or by implementation specific means.

179.9.4.3 Level separation mismatch ratio

Level separation mismatch ratio (R_{LM}) is defined using the method in 120D.3.1.2.

The R_{LM} (min) requirement in Table 179–7 shall be met when the transmitter equalization is set to each of the initial conditions defined in Table 179–8.

179.9.4.4 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is determined using the method in 120D.3.1.7 with the following exceptions:

- The linear fit pulse response $p(k)$ is determined using the linear fit procedure in 179.9.4.2.1.
- The continuous time filter parameters are provided in Table 179–24.
- For calculation of SNR_{ISI} using Equation (120D–8) a value of 20 is used for N_b .
- A time offset is added to t_p whose value is swept from -0.5 UI to 0.5 UI when calculating $ISI_{cursors}$. SNR_{ISI} is defined as the lowest value found across the time offset sweep.
- The transmitter equalizer setting is chosen, within the required coefficient ranges (see 179.9.4.2.5), to result in the highest SNR_{ISI} value.

179.9.4.5 Host channel insertion loss

The recommended insertion loss for host channels is provided in 179A.4.

179.9.4.6 Signal-to-noise-and-distortion ratio

Signal-to-noise-and-distortion ratio (*SNDR*) is defined for a specific transmit equalization setting and calculated using the following procedure.

Set the transmitters on all lanes to the same transmit equalization setting.

Set the transmitter on the lane under test to transmit PRBS13Q and the transmitters on all other lanes to transmitting either PRBS31Q or PCS data.

Capture the output of the lane under test and compute the linear fit pulse response, $p(k)$, and the linear fit error, $e(k)$, according to 179.9.4.2.1. The standard deviation of $e(k)$ is denoted by σ_e .

Measure the RMS deviation from the mean voltage at a fixed low-slope point in runs of at least 6 consecutive identical PAM4 symbols. PRBS13Q includes such a run for each of the PAM4 levels. The average of the four measurements is denoted by σ_n .

The signal power P_{signal} is defined by Equation (179–6) and Equation (179–7).

$$m_0 = 1 + (k_{peak} - 1) \bmod M \quad (179-6)$$

$$P_{Signal} = \sum_{i=0}^{N_p-1} p(M \times i + m_0)^2 \quad (179-7)$$

where

k_{peak} is the value of k that maximizes $p(k)$
 M and N_p are defined in 179.9.4.2.1.

SNDR is defined by Equation (179–8).

$$SNDR = 10 \log_{10} \left(\frac{P_{Signal}}{\sigma_e^2 + \sigma_n^2} \right) \quad (179-8)$$

When *SNDR* is measured at any of the initial condition settings defined in Table 179–8, its value shall be equal to or larger than the minimum value specified in Table 179–9 for that setting.

179.9.4.7 Output jitter

Output jitter is characterized by three parameters: JH_{RMS} , JH_{4u} , and EOJ_{03} . These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the host channel. The equalizer setting is chosen to minimize any or all of the jitter parameters.

Jitter is measured separately on each lane. The lane under test transmits either PRBS13Q or PRBS9Q (defined in 120.5.11.2.a). Lanes not under test transmit either PRBS31Q or scrambled idle. For EOJ_{03} measurements, transmitter outputs of the lanes not under test are disabled, but other circuitry in lanes not under test should be kept active during the measurement.

Table 179–9—SNDR limits

ic_req	SNDR (min)	Units
PRESET 1	33.5	dB
PRESET 2	27.5	dB
PRESET 3	30.7	dB
PRESET 4	30.2	dB
PRESET 5	28.7	dB
PRESET 6 / INITIALIZE ^a	31	dB

^a The INITIALIZE setting is identical to PRESET 6.

NOTE—Outputs of lanes not under test are disabled in order to minimize the effect of crosstalk on the measurement.

Measurement is performed with a clock recovery unit (CRU) with a corner frequency of 4 MHz and a slope of 20 dB/decade.

179.9.4.7.1 JH_{RMS}

Calculation of the jitter parameter JH_{RMS} uses the timing of multiple transitions in the pattern to estimate the jitter caused by clock phase noise while minimizing the effect of additive noise, using the method described in this subclause or equivalent methods.

Select a set A of transitions from the test pattern used in the test. The size of this set should be large enough to enable calculation of JH_{RMS} (as defined below) with sufficient accuracy. The set A should include multiple transitions between different PAM4 levels. The location of the specific transitions in the test pattern and the threshold levels at which the transition times are extracted are chosen from the measured signal to minimize JH_{RMS}.

For each specific transition $i \in A$:

- Obtain a set $A_i = \{t_i(1), t_i(2), \dots\}$ of transition times modulo the period of the pattern. The size of this set should be large enough to enable estimating the mean and standard deviation with sufficient accuracy.
- Estimate the mean μ_i and standard deviation σ_i from the measured dataset A_i .
- Measure the slope of the signal at time μ_i , denoted s_i .

From the collections of σ_i and s_i , calculate the best-fit coefficients for the polynomial in $1/s_i^2$ defined by Equation (179–9):

$$\sigma_i^2 = a + b \frac{1}{s_i^2} + c \frac{1}{s_i^4} \quad (179-9)$$

The free coefficient a is the extrapolation of the measured data to an infinite slope. JH_{RMS} is defined as the square root of a .

179.9.4.7.2 JH4u

Calculation of the jitter parameter JH4u uses the timing of multiple transitions in the pattern to estimate the jitter caused by clock phase noise while minimizing the effect of additive noise, using the method described in this subclause or equivalent methods.

Select a set A of transitions from the test pattern used in the test. The size of this set should be large enough to enable calculation of JH4u (as defined below) with sufficient accuracy. The set A should include multiple transitions between different PAM4 levels. The location of the specific transitions in the test pattern and the threshold levels at which the transition times are extracted are chosen from the measured signal to minimize JH4u. For example, locations where the transition slope is low due to ISI should be eliminated.

For each specific transition $i \in A$:

- Obtain a set $A_i = \{t_i(1), t_i(2), \dots\}$ of transition times modulo the period of the pattern. The size of this set should be large enough to enable calculation of μ_i , $RJrms_i$, and $DJdd_i$ (as defined below) with sufficient accuracy.
- Estimate the mean μ_i from the measured dataset A_i .
- Using the dual-Dirac jitter decomposition method, estimate the random jitter RMS component $RJrms_i$ and the deterministic jitter component $DJdd_i$ from the measured dataset A_i .
- Measure the slope of the signal at time μ_i , denoted s_i .

From the collections of $RJrms_i$ and s_i , calculate the best-fit coefficients for the polynomial in $1/s_i^2$ defined by Equation (179–10):

$$RJrms_i^2 = a_{RJ} + b_{RJ} \frac{1}{s_i^2} + c_{RJ} \frac{1}{s_i^4} \quad (179-10)$$

The free coefficient a_{RJ} is the extrapolation of the measured data to an infinite slope. $RJrms$ is defined as the square root of a_{RJ} .

From the collections of $DJdd_i$ and s_i , calculate the best-fit coefficients for the polynomial in $1/|s_i|$ defined by Equation (179–11):

$$DJdd_i = a_{DJ} + b_{DJ} \frac{1}{|s_i|} \quad (179-11)$$

The free coefficient a_{DJ} is the extrapolation of the measured data to an infinite slope. $DJdd$ is defined as a_{DJ} .

JH4u is defined by Equation (179–12).

$$JH4u = DJdd + 2 Q_4 RJrms \quad (179-12)$$

Where $Q_4 = 3.891$.

NOTE— Q_4 is an approximated solution of $Q(Q_4) = 5 \times 10^{-5}$, where the Q function is defined in Equation (95–1).

179.9.4.7.3 EOJ₀₃

EOJ₀₃ is calculated from measurements using the method specified in 162.9.4.7 for even-odd jitter. Only the transitions R03 and F30 defined in 179.9.4.7.2 are used. The measured values for these two transitions are denoted EOJ₁ and EOJ₂.

EOJ₀₃ is defined by Equation (179–13).

$$EOJ_{03} = \sqrt{\frac{EOJ_1^2 + EOJ_2^2}{2}} \quad (179-13)$$

179.9.4.8 Transmitter return loss specifications

179.9.4.8.1 Transmitter effective return loss (ERL)

The ERL of the transmitter at TP2 is defined by the procedure in 93A.5 using the values in Table 179–10 and Table 179–22, with differential reference impedance equal to 92.5 Ω, and with T_{fx} equal to twice the test fixture delay. The test fixture delay is defined as the propagation delay between the coaxial connector on the test fixture and the host-facing connection, excluding the discontinuity at the HCB-to-host mating interface. T_{fx} is provided by the test fixture provider.

Table 179–10—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	1600	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—
Number of samples per unit interval	M	32	—

Transmitter ERL at TP2 shall meet the requirement in Table 179–7.

179.9.4.8.2 Transmitter modal ERL

Transmitter modal ERL (ERL_{CC} or ERL_{DC}) is the ERL computed according to 93A.5 using the values in Table 179–10 and Table 179–24 and T_{fx} as defined in 179.9.4.8.1, with the following exceptions.

- For transmitter ERL_{CC} , $s_{ii}(f)$ (see 93A.5.1) is replaced with the transmitter $s_{CC22}(f)$.
- For transmitter ERL_{DC} , $s_{ii}(f)$ is replaced with the transmitter $s_{DC22}(f)$.

$s_{CC22}(f)$ and $s_{DC22}(f)$ are defined in 176D.8.3.3.

179.9.4.8.3 Transmitter modal return loss specifications

The transmitter modal return loss specifications are common-mode to common-mode and common-mode to differential-mode.

The transmitter common-mode to common-mode return loss specification is satisfied by either meeting the requirement of Equation (179–14) or having ERL_{CC} higher than the value in Table 179–11.

The transmitter common-mode to differential-mode return loss specification is satisfied by either meeting the requirement of Equation (179–15) or having ERL_{DC} higher than the value in Table 179–11.

Table 179–11—Transmitter modal return loss specifications

Specification	RL mask equation	RL mask figure	Modal ERL type	Modal ERL minimum (dB)
Common-mode to common-mode	Equation (179–14)	Figure 179–4	ERL _{CC}	2
Common-mode to differential-mode	Equation (179–15)	Figure 179–5	ERL _{DC}	17

$$RL_{cc}(f) \geq \begin{cases} 2 & 0.05 \leq f < 4 \\ 2 + \frac{2}{40}(f - 4) & 4 \leq f < 44 \\ 4 + \frac{2}{16}(44 - f) & 44 \leq f < 60 \\ 2 & 60 \leq f \leq 67 \end{cases} \quad (179-14)$$

where

$RL_{cc}(f)$ is the negative of the value in dB of $s_{CC22}(f)$ (defined in 176D.8.3.3)
 f is the frequency in GHz

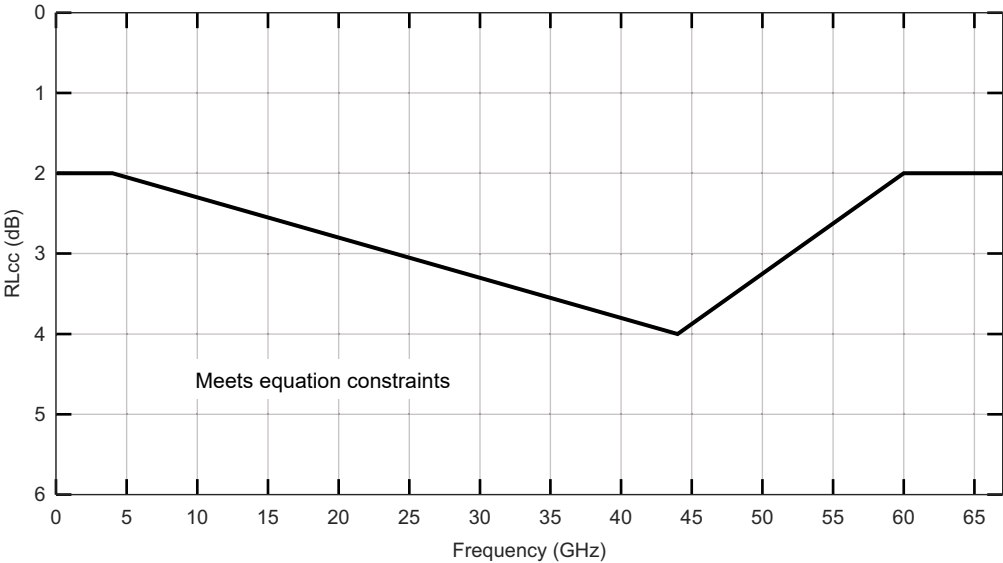


Figure 179–4—Common-mode to common-mode return loss mask

$$RL_{dc}(f) \geq \begin{cases} 23 - 22 \frac{f}{106.25} & 0.05 \leq f < 53.125 \\ 12 & 53.125 \leq f < 67 \end{cases} \quad (179-15)$$

where

$RL_{dc}(f)$ is the negative of the value in dB of $s_{DC22}(f)$ (defined in 176D.8.3.3)
 f is the frequency in GHz

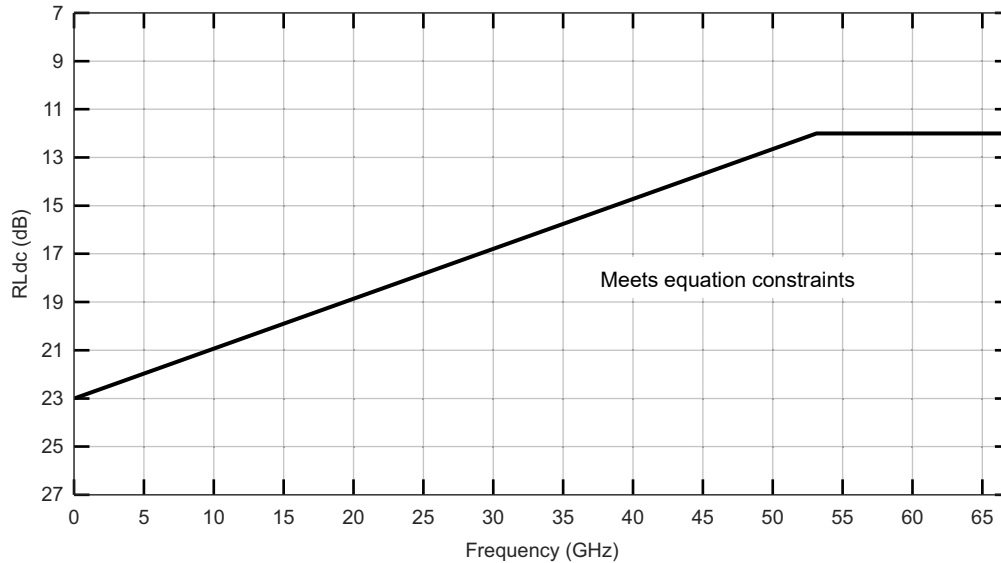


Figure 179-5—Common-mode to differential-mode return loss mask

179.9.5 Receiver characteristics

The receiver on each lane shall meet the specifications given in Table 179-12 and detailed in the referenced subclauses. Unless specified otherwise, all receiver measurements are made for each lane separately, at TP3, utilizing the test fixtures specified in Annex 179B.

Recommended receiver characteristics at TP5d are provided in Annex 179A.3.

Table 179-12—Summary of receiver specifications at TP3

Parameter	Reference	Value	Units
Signaling rate (range)	179.9.5.1	106.25 ± 50 ppm	GBd
Transmitter output amplitude tolerance	179.9.5.5	0.5	V
Interference tolerance	179.9.5.3	Table 179-15	—
Jitter tolerance	179.9.5.4	Table 179-16	—
Effective return loss, ERL (min)	179.9.5.6.1	7.3	dB
Differential-mode to common-mode return loss	179.9.5.6.3	Table 179-17	—

179.9.5.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 179.9.5.5, 179.9.5.3, and 179.9.5.4 for any signaling rate in the range specified in Table 179–12.

179.9.5.2 Error ratio requirements for receiver tests

A receiver on each lane is required to meet the expected BLER specified in 179.2, using the test method in either 174A.9.5, 174A.9.7, or 174A.9.6 with parameters provided in Table 179–13. The error mask $H_{\max}(k)$ to be used in the method of 174A.9.5 is provided in Table 179–14.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m^{(i)}(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m^{(i)}(k))$ for $k = n-2$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $[\log_{10}(H_m^{(i)}(k)) - \log_{10}(H_m^{(i)}(4))] / (k - 4)$ should be less than $[\log_{10}(H_m^{(i)}(4)) - \log_{10}(H_m^{(i)}(2))] / 2$ for each k greater than 4.

Table 179–13—Error ratio parameters

Parameter name	Value
p	
200GBASE-CR1	1
400GBASE-CR2	2
800GBASE-CR4	4
1.6TBASE-CR8	8
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	1.6×10^{-5}
BER	2.76×10^{-4}

Table 179–14—Receiver error mask

Test symbol errors per test block, k (see 174A.9.5)	$H_{\max}(k)$			
	$p = 1$	$p = 2$	$p = 4$	$p = 8$
1	3.3×10^{-1}	3.5×10^{-1}	2.6×10^{-1}	1.6×10^{-1}
2	2.5×10^{-1}	1.3×10^{-1}	4.8×10^{-2}	1.4×10^{-2}
3	1.3×10^{-1}	3.3×10^{-2}	6×10^{-3}	8.8×10^{-4}
4	4.7×10^{-2}	6.2×10^{-3}	5.5×10^{-4}	3.9×10^{-5}
5	1.4×10^{-2}	9.1×10^{-4}	4×10^{-5}	1.4×10^{-6}
6	3.5×10^{-3}	1.1×10^{-4}	2.4×10^{-6}	4.1×10^{-8}
7	7.4×10^{-4}	1.2×10^{-5}	1.2×10^{-7}	9.9×10^{-10}
8	1.4×10^{-4}	1.1×10^{-6}	5.5×10^{-9}	2.1×10^{-11}

Table 179–14—Receiver error mask (*continued*)

Test symbol errors per test block, k (see 174A.9.5)	$H_{\max}(k)$			
	$p = 1$	$p = 2$	$p = 4$	$p = 8$
9	2.3×10^{-5}	8.8×10^{-8}	2.2×10^{-10}	3.9×10^{-13}
10	3.4×10^{-6}	6.4×10^{-9}	7.6×10^{-12}	6.3×10^{-15}
11	4.5×10^{-7}	4.2×10^{-10}	2.4×10^{-13}	9.2×10^{-17}
12	5.5×10^{-8}	2.5×10^{-11}	7×10^{-15}	1.2×10^{-18}
13	6.3×10^{-9}	1.4×10^{-12}	1.8×10^{-16}	1.4×10^{-20}
14	6.6×10^{-10}	7.2×10^{-14}	4.5×10^{-18}	1.6×10^{-22}
15	6.4×10^{-11}	3.4×10^{-15}	1×10^{-19}	1.6×10^{-24}
16	5.9×10^{-12}	1.5×10^{-16}	2.1×10^{-21}	1.4×10^{-26}

179.9.5.3 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure described in 179.9.5.3.1 through 179.9.5.3.5.

Receiver interference tolerance is specified by two tests defined in Table 179–15. Test L includes a low-loss channel. Test H includes a high-loss channel.

NOTE—The channel for Test H is dependent on the host class that a PMD under test complies with.

A receiver shall meet the requirements of this test under both Test L and Test H conditions.

Table 179–15—Interference tolerance test parameters

Parameter	Test L (low loss)		Test H (high loss)		Units
	Min	Max	Min	Max	
Test pattern	PRBS31Q				
Test channel insertion loss at 53.125 GHz ^a					
Host class HL	15.5	16.5	36.5	37.5	dB
Host class HN	15.5	16.5	31.5	32.5	dB
Host class HH	15.5	16.5	26.5	27.5	dB
Cable assembly insertion loss at 53.125 GHz					
Host class HL	15.5	16.5	33.5	34.5	dB
Host class HN	15.5	16.5	28.5	29.5	dB
Host class HH	15.5	16.5	23.5	24.5	dB
COM ^b	3		3		dB

^a Insertion loss between the two test references (see Figure 110–3b).

^b COM is calculated as defined in 179.9.5.3.3. Meeting the test requirements with a lower value of COM demonstrates margin to the specification but is not required for compliance.

179.9.5.3.1 Test setup

The interference tolerance test is performed with the setup similar to the one described in 110.8.4.2.1. The test setup includes noise injection for all lanes of the pattern generator.

The cable assembly used in the test channel (see 110.8.4.2.2) shall meet the cable assembly Channel Operating Margin (COM) as specified in 179.11.4.

179.9.5.3.2 Test channel

The test channel is the same as the one defined in 110.8.4.2.2, except that the cable assembly meets the requirements of 179.11, the test channel loss meets the requirements of Table 179–15, and the cable assembly test fixture meets the requirements of 179B.3.

179.9.5.3.3 Test channel calibration

The scattering parameters of the test channel are measured at the test references as illustrated in Figure 110–3b using the cable assembly test fixtures specified in 179B.3.

The insertion loss at 53.125 GHz of the signal path between the test references in Figure 110–3b is within the limits in Table 179–15.

The COM is calculated using the method and parameters of 179.11.4 with the following considerations:

- a) The test channel is measured between the Tx and Rx test references shown in Figure 110–3b. COM is calculated using the receiver partial host channel, package, and device termination models in Table 179–22 corresponding to the host class of the receiver under test.
- b) The transition time T_r is measured at the Tx test reference with the transmit equalizer set to PRESET 1 (i.e., equalization off with maximum amplitude), using the method in 120E.3.1.5, with an exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz.
- c) EOJ_{03} , JH4u, and JH_{RMS} are measured at the Tx test reference with pattern generator noise injection disabled (see 179.9.5.3.4). The measured values comply with the specification in Table 179–7. In the calculation of COM, A_{DD} and σ_{RJ} are calculated from the measured values of JH4u and JH_{RMS} using Equation (179–16) and Equation (179–17), replacing the values in Table 179–24. It is recommended to adjust the pattern generator jitter such that JH4u and JH_{RMS} are as close as practical to their limits in Table 179–7.
- d) In the calculation of the channel S-parameters in 178A.1.2, the transmitter S-parameter model $S^{(t)}$ is omitted. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A–13) uses a rise time filter $H_t(f)$ (see 178A.1.6.2) calculated with T_r measured in step b.
- e) In the calculation of COM, SNR_{TX} is set to the SNDR value with $N_p=85$ measured at the Tx test reference with pattern generator noise disabled.
- f) The interference noise (see 178A.1.7.5) is derived from the S-parameters of the test channel. For a given RMS noise voltage σ_{ns} measured at the Tx test reference, the spectral density of the noise source $N_s(f)$ used for calculation of COM may be calculated using Equation (179–18). Alternatively, $N_s(f)$ may be obtained by measurement of the noise spectral density at the output of the noise source.
- g) The value of σ_{ns} that yields the target value of COM specified in Table 179–15 is determined (e.g., via iterative calculation of COM). The noise level at the Tx test reference is then configured so that the measured RMS voltage matches σ_{ns} .

$$A_{DD} = \frac{\frac{JH4u}{2} + Q4d \sqrt{(Q4d^2 + 1) \times JH_{RMS}^2 - \left(\frac{JH4u}{2}\right)^2}}{Q4d^2 + 1} \quad (179-16)$$

$$\sigma_{RJ} = \frac{\frac{JH4u}{2} - A_{DD}}{Q4d} \quad (179-17)$$

where $Q4d = 3.719$.

NOTE 1— $Q4d$ is an approximated solution of $Q(Q4d) = 10^{-4}$, where the Q function is defined in Equation (95-1).

NOTE 2—The calculation of A_{DD} can, under certain conditions, result in a negative argument of the square root. If this condition occurs, the recommended solution is to increase sinusoidal jitter or bounded uncorrelated jitter such that the argument is positive.

$$N_s(f) = \begin{cases} \left(\frac{2\sigma_{ns}^2}{\beta \cdot f_b}\right) \frac{(f/f_{hp})^2}{1 + (f/f_{hp})^2} & 0 \leq f \leq \frac{f_b}{2} \\ 0 & \text{otherwise} \end{cases} \quad (179-18)$$

where

$$\begin{aligned} f_b & \text{ is defined in Table 179-24} \\ f_{hp} & \text{ is 6 GHz} \\ \beta & \text{ is } 1 - (2f_{hp}/f_b) \operatorname{atan}\left(\frac{f_b}{2f_{hp}}\right). \end{aligned}$$

179.9.5.3.4 Pattern generator and noise injection

The pattern generator transmits data to all lanes of the device under test. At the start of transmitter training, the pattern generator output amplitude is set such that steady-state voltage v_f (see 179.9.4.2.2) at the Tx test reference is the minimum specified in Table 179-7. v_f does not exceed that minimum during the test. The pattern generator equalization is set to the INITIALIZE setting in Table 179-8. During the test, the pattern generator equalization does not exceed the coefficient range in 179.9.4.2.5.

NOTE—The pattern generator requirements imply that the differential peak-to-peak output voltage (as defined in 176D.8.2) at the pattern generator output does not exceed 0.8 V.

Noise is added to the signal before the Tx test reference, with noise level set according to step g) in 179.9.5.3.3.

The noise required for each lane is calibrated. The noise may be added either to one lane at a time or using multiple noise sources to all lanes at the same time.

To avoid excessive low-frequency weighting to the receiver input noise, the noise added to the signal is bounded by the normalized spectral density mask defined in Equation (179-19), Equation (179-20), and Equation (179-21) with $f_1 = 8$ GHz and $f_2 = 5$ GHz, and illustrated in Figure 179-6.

$$NSD_{avg} = \frac{1}{(f_b/2 - f_1)} \int_{f_1}^{f_b/2} NSD(f) df \quad (179-19)$$

$$10 \log_{10} \left(\frac{NSD(f)}{NSD_{avg}} \right) \geq -3 + 3.6(f/f_b) \quad f_1 \leq f \leq f_b/2 \quad (179-20)$$

$$10\log_{10}\left(\frac{NSD(f)}{NSD_{avg}}\right) \leq \begin{cases} -12 + 15f/f_2 & 0 \leq f < f_2 \\ 3 & f_2 \leq f \leq f_b/2 \end{cases} \quad (179-21)$$

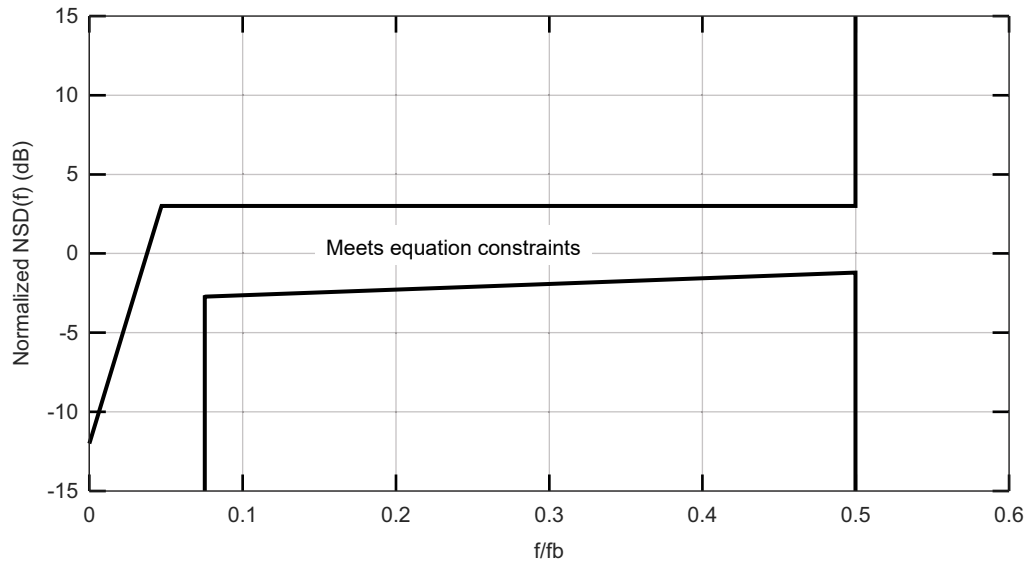


Figure 179-6—NSD(f) constraints

179.9.5.3.5 Test procedure

The pattern generator is first configured to transmit the training pattern defined in 179.8.9. During this initialization period, the device under test (DUT) configures the pattern generator transmit equalizer coefficients and precoding to the settings it would select using the start-up protocol described in 179.8.9 and the receiver is tuned using its optimization method. The settings may be communicated via the start-up protocol or by other means.

After the pattern generator has been configured and the receiver tuned, the pattern generator is set to generate the test pattern specified in Table 179-15. During the test, transmitters in all lanes of the PMD under test transmit either scrambled idle or PRBS31Q, with equalization turned off (PRESET 1 condition).

A receiver shall meet the error ratio requirements defined in 179.9.5.2 under the conditions specified for this test for the host class that the PMD complies with, for both Test L and Test H in Table 179-15.

179.9.5.4 Receiver jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 179.9.5.4.1 and 179.9.5.4.2.

179.9.5.4.1 Test setup

Jitter tolerance is measured with a channel meeting the insertion loss of Test H as specified in Table 179-15. The pattern generator includes sinusoidal jitter injection and complies with the output jitter specifications for Host class HL in Table 179-7.

179.9.5.4.2 Test procedure

The jitter tolerance test procedure is similar to that of 179.9.5.3, with the following exceptions:

- For each case in Table 179–16, jitter with the specified frequency is applied to the pattern generator and the jitter amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency at the Tx test reference (see Figure 110–3a).
- The test channel calibration (see 179.9.5.3.3) is performed with the transmitter output set to the jitter frequency and amplitude for Case G of Table 179–16. The additive noise level obtained is used for all other cases.

A receiver shall meet the error ratio requirement defined in 179.9.5.2 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 179–16.

Table 179–16—Receiver jitter tolerance parameters

Parameter	Case A	Case B	Case C	Case D	Case E	Case F	Case G	Units
Jitter frequency	0.04	0.1333	0.4	1.333	4	12	40	MHz
Jitter amplitude (peak-to-peak)	5	1.5	0.5	0.15	0.05	0.05	0.05	UI

NOTE—The A_{DD} (Equation (179–16)) and σ_{RJ} (Equation (179–17)) calculated from transmitter measurements in this test can be higher than the values in Table 179–24. A suitable channel should be chosen in order to meet the COM requirement with these values.

179.9.5.5 Transmitter output amplitude tolerance

Transmitter output amplitude tolerance of a receiver is defined as the maximum transmitter steady-state voltage that the receiver can tolerate, using the Test L (low loss) test channel specified in 179.9.5.3.2, such that it satisfies the error ratio requirements defined in 179.9.5.2 when the PMD transmit function operates in DATA mode (see 179.8.2).

The transmitter steady-state voltage is measured as specified in 179.9.4.2.2 at the output of the pattern generator used in the test. The differential peak-to-peak output voltage of the pattern generator does not exceed the maximum value specified in Table 179–7.

The pattern generator is initially configured to transmit training frames as defined in 178B.7.3 with transmit equalization at the INITIALIZE setting (see Table 179–8). During the initialization period, the device under test (DUT) may further configure the pattern generator to the transmit equalizer and precoding settings it would select using the ILT function (see 179.8.9). The settings may be communicated via the ILT function or by other means. After the pattern generator has been configured, the BLER is measured as specified in 174A.9 or 174A.11.

NOTE—Steady-state voltage is defined with PRESET 1. It is not initially generated by a transmitter, due to the INITIALIZE setting in Table 179–8. The receiver is not required to tolerate PRESET 1 unless it specifically requests it. The voltage observed at the receiver input is further attenuated by the test channel. Therefore, the receiver is not required to tolerate the maximum transmitter output voltage at its input.

179.9.5.6 Receiver return loss specifications

179.9.5.6.1 Receiver ERL

The ERL of the receiver at TP3 is computed using the procedure in 93A.5 with the values in Table 179–10 and Table 179–22, with differential reference impedance equal to $92.5\ \Omega$, and with T_{fx} equal to twice the test fixture delay. The test fixture delay is defined as the propagation delay between the coaxial connector on the test fixture and the host-facing connection, excluding the discontinuity at the HCB-to-host mating interface. T_{fx} is provided by the test fixture provider.

Receiver ERL at TP3 shall meet the requirement specified in Table 179–12.

179.9.5.6.2 Receiver modal ERL

Receiver modal ERL (ERL_{CD}) is the effective return loss computed according to 93A.5 using the values in Table 179–10 and Table 179–24 and T_{fx} as defined in 179.9.5.6.1, with the exception that $s_{ii}(f)$ (see 93A.5.1) is replaced with the receiver $s_{CD11}(f)$, which is defined in 176D.8.3.3.

179.9.5.6.3 Receiver modal return loss specifications

The receiver modal return loss specification is differential-mode to common-mode. It is satisfied by either meeting the requirement of Equation (179–22) or having ERL_{CD} value higher than the value in Table 179–17.

Table 179–17—Receiver modal return loss specifications

Specification	RL mask equation	RL mask figure	Modal ERL type	Modal ERL minimum (dB)
Differential-mode to common-mode	Equation (179–22)	Figure 179–7	ERL_{CD}	17

$$RLcd(f) \geq \begin{cases} 23 - 22 \frac{f}{106.25} & 0.01 \leq f < 53.125 \\ 12 & 53.125 \leq f < 67 \end{cases} \quad (179-22)$$

where

$RLcd(f)$ is the negative of the value in dB of $s_{CD11}(f)$ (defined in 176D.8.3.3)
 f is the frequency in GHz

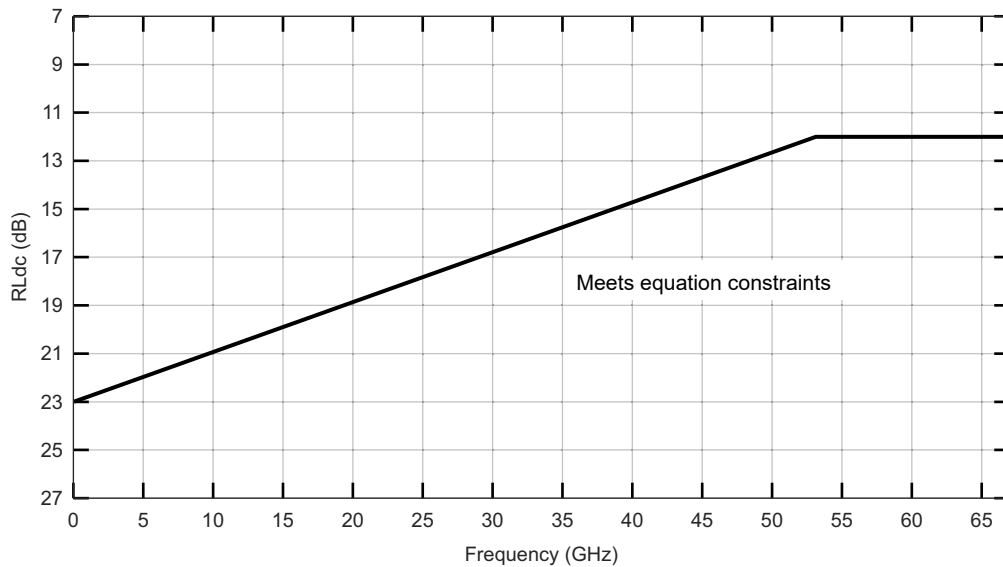


Figure 179-7—Differential-mode to common-mode return loss mask

179.10 Channel (TP0d-TP5d) characteristics

The channel is defined between TP0d and TP5d to include the transmitter and receiver differential controlled impedance package and PCB and the cable assembly as illustrated in Figure 179-2. The channel insertion loss, return loss, COM, and the host channel parameters are provided informatively in 179A.4 through 179A.7.

Channel definitions apply for links between two PHYs of the same type: 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8.

179.11 Cable assembly characteristics

Cable assemblies defined in this subclause contain insulated conductors terminated in a connector at each end for use as link segments between MDIs. Cable assemblies are primarily intended as point-to-point links between 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8 PHYs using controlled impedance cables.

Four cable assembly types are specified:

- 200GBASE-CR1: Cable assembly that supports single-lane links between two 200GBASE-CR1 PHYs.
- 400GBASE-CR2: Cable assembly that supports two-lane links between two 400GBASE-CR2 PHYs.
- 800GBASE-CR4: Cable assembly that supports four-lane links between two 800GBASE-CR4 PHYs.
- 1.6TBASE-CR8: Cable assembly that supports eight-lane links between two 1.6TBASE-CR8 PHYs.

The specifications for the four cable assembly types are identical except the number of lanes.

For each of the cable assembly types, four cable assembly classes are defined, labeled CA-A, CA-B, CA-C, and CA-D. The cable assembly classes differ by the maximum insertion loss (see 179.11.2).

The cable assembly types are described in Annex 179D. The MDIs are defined in Annex 179C.

Within the cable assembly, all lanes shall be AC-coupled. It is recommended that AC-coupling is within the plug connectors. The low-frequency 3 dB cutoff of the AC-coupling shall be less than 250 kHz. If AC-coupling is implemented using coupling capacitors, it is recommended that the value of the capacitors be less than or equal to 100 nF, to limit the inrush charge.

NOTE—There are various methods for AC-coupling in actual implementations.

All cable assembly measurements are to be made between TP1 and TP4 with cable assembly test fixtures as specified in Annex 179B. These cable assembly specifications are based upon twinaxial cable characteristics, but other cable types are acceptable if the specifications of this subclause are met.

Table 179–18 provides a summary of the cable assembly characteristics and references to the subclauses addressing each parameter.

Table 179–18—Cable assembly characteristics summary

Description	Reference	Value	Unit
Fitted insertion loss at 53.125 GHz, $ILdd_{fit}$ (max)	179.11.2		
CA-A		19	dB
CA-B		24	dB
CA-C		29	dB
CA-D		34	dB
Fitted insertion loss at 53.125 GHz, $ILdd_{fit}$ (min)	179.11.2	16	dB
Minimum cable assembly ERL ^a	179.11.3.1	8.25	dB
Differential-mode to common-mode return loss	179.11.3.3	Table 179–20	—
Common-mode to common-mode return loss	179.11.3.3	Table 179–20	—
Minimum COM	179.11.4	3	dB
Channel signal to common-mode ratio, $SCMR_{CH}$ (min)	179.11.5	9	dB
NOTE—The expected cable assembly reach is 0.5 m for CA-A, 1 m for CA-B, 1.5 m for CA-C, and 2 m for CA-D. Compliant cable assemblies may be longer. The length of a cable assembly does not imply compliance to specifications.			

^a Cable assemblies with a COM greater than 4 dB are not required to meet minimum ERL.

179.11.1 Reference impedance

The reference differential impedance for cable assembly specifications is 92.5 Ω . The reference common-mode impedance for cable assembly specifications is 23.125 Ω .

179.11.2 Cable assembly fitted insertion loss

The fitted insertion loss at 53.125 GHz of a cable assembly shall be less than or equal to $ILdd_{fit}$ (max) in Table 179–18 for the cable assembly class the cable adheres to.

The fitted insertion loss at 53.125 GHz of a cable assembly shall be greater than or equal to $IL_{dd_{fit}}$ (min) in Table 179–18.

Fitted insertion loss is calculated using the procedure in 93A.3 with $f_{min} = 0.01$ GHz, $f_{max} = 67$ GHz, and $\Delta f = 0.01$ GHz.

179.11.3 Cable assembly return loss specifications

179.11.3.1 Cable assembly ERL

The cable assembly ERL at TP1 and at TP4 is defined by the procedure in 93A.5 with the values in Table 179–19 and Table 179–22, with differential reference impedance equal to 92.5Ω , and with T_{fx} equal to twice the test fixture delay. The test fixture delay is defined as the propagation delay between the coaxial connector on the test fixture and the first discontinuity of the MDI connector. T_{fx} is provided by the test fixture provider.

The values of cable assembly ERL at TP1 and at TP4 shall meet the requirement specified in Table 179–18 for cable assemblies that have a COM less than 4 dB.

Table 179–19—Cable assembly ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	9000	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—
Number of samples per unit interval	M	32	—

179.11.3.2 Cable assembly modal ERL

Cable assembly modal ERL (ERL_{CC} or ERL_{CD}) is the ERL computed according to 93A.5 using the values in Table 179–19 and Table 179–24 and T_{fx} as defined in 179.11.3.1, with the following exceptions.

- For cable assembly input ERL_{CC} , $s_{ii}(f)$ (see 93A.5.1) is replaced with the cable assembly $s_{CC11}(f)$.
- For cable assembly input ERL_{CD} , $s_{ii}(f)$ is replaced with the cable assembly $s_{CD11}(f)$.
- For cable assembly output ERL_{CC} , $s_{ii}(f)$ is replaced with the cable assembly $s_{CC22}(f)$.
- For cable assembly output ERL_{CD} , $s_{ii}(f)$ is replaced with the cable assembly $s_{CD22}(f)$.

$s_{CC11}(f)$, $s_{CD11}(f)$, $s_{CC22}(f)$, and $s_{CD22}(f)$ are defined in 178A.1.3.

179.11.3.3 Cable assembly modal return loss specifications

The cable assembly modal return loss specifications are common-mode to common-mode and differential-mode to common-mode.

The cable assembly common-mode to common-mode return loss specification is satisfied by either meeting the requirement of Equation (179–14) or having ERL_{CC} higher than the value in Table 179–20.

The cable assembly differential-mode to common-mode return loss specification is satisfied by either meeting the requirement of Equation (179–15) or having ERL_{CD} higher than the value in Table 179–20.

Table 179–20—Cable assembly modal return loss specifications

Specification	RL mask equation	RL mask figure	Modal ERL type	Modal ERL minimum (dB)
Common-mode to common-mode	Equation (179–14)	Figure 179–4	ERL_{CC}	2
Differential-mode to common-mode	Equation (179–22)	Figure 179–7	ERL_{CD}	17

179.11.4 Cable assembly Channel Operating Margin

The cable assembly Channel Operating Margin (COM) for each lane is derived from measurements of the cable assembly signal, near-end crosstalk, and far-end crosstalk paths. COM is computed using the procedure in 178A.1 and parameter values provided in 179.11.4.1. The signal paths for calculation of COM are defined in 179.11.4.2.

For each cable assembly class, the valid combinations of transmitter host class and receiver host class are listed in Table 179–21. For each valid combination., COM for any lane within the cable assembly shall be greater than or equal to the minimum COM specified in Table 179–18.

Table 179–21—Cable assembly class and host class valid combinations

Cable assembly class	Host classes, transmitter side	Host classes, receiver side	Number of combinations
CA-A	HN or HL	HL, HN, or HH	6
	HH	HL or HN	2
CA-B	HL	HL, HN, or HH	3
	HN	HL or HN	2
	HH	HL	1
CA-C	HL	HL or HN	2
	HN	HL	1
CA-D	HL	HL	1

179.11.4.1 COM parameters

The parameters in Table 179–22 and Table 179–23 represent a partial host channel that includes a combination of a packaged device and a PCB, using the S-parameter model defined in 178A.1.4 and 178A.1.5. This model is not a host channel specification.

The parameters in Table 179–24 correspond to behavioral models of transmitters and receivers that, with suitable host channel models, are compliant to the PMD specifications in this clause. The purpose of these parameters is to compute COM, a channel metric, and they do not represent requirements for transmitter and receiver implementations.

It is expected that the PMD specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation and host channel construction.

Table 179–22—Host model parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF
Device package model, class A			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length	$z_p^{(1)}$	See Table 179–23	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Device package model, class B			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length	$z_p^{(1)}$	See Table 179–23	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF

Table 179–22—Host model parameters (*continued*)

Parameter	Symbol	Value	Units
Partial host channel model			
Single-ended board capacitance at the package-to-board interface	C_0	0	nF
Transmission line parameter γ_0	$\gamma_0^{(h)}$	0	1/mm
Transmission line parameter a_1	$a_1^{(h)}$	5.95×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	$a_2^{(h)}$	2.6×10^{-5}	ns/mm
Transmission line parameter τ	$\tau^{(h)}$	5.79×10^{-3}	ns/mm
Transmission line characteristic impedance	$Z_c^{(h)}$	92.5	Ω
Transmission line length	$z_p^{(h)}$	See Table 179–23	mm
Single-ended capacitance at the model-to-measurement interface	C_1	0	nF
Single-ended reference resistance	R_0	46.25	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package and partial host channel model parameters are based on the reference impedance R_0 .			

Table 179–23—Partial host channel model parameters per Host class

Parameter	Host class			Units
	HL	HN	HH	
Device package model class	A	B	B	—
Package transmission line 1 length, $z_p^{(1)}$	8	15	45	mm
Partial host PCB transmission line length, $z_p^{(h)}$	22	83	74	mm

NOTE—For each host class, the sum of the differential insertion loss (ILdd) at 53.125 GHz of the partial host channel (excluding the device termination) and the reference cable assembly test fixture (see Equation (179B–2) and Figure 179A–1) is equal to the recommended maximum host channel insertion loss in 179A.4 for that host class.

Table 179–24—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$	0	—

Table 179–24—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, coefficient –2	$c(-2)$		
Minimum value		0	
Maximum value		0.14	—
Step size		0.02	
Transmitter equalizer, coefficient –1	$c(-1)$		
Minimum value		–0.34	
Maximum value		0	—
Step size		0.02	
Transmitter equalizer, coefficient 0	$c(0)$		
Minimum value		0.54	—
Transmitter equalizer, coefficient 1	$c(1)$		
Minimum value		–0.2	
Maximum value		0	—
Step size		0.02	
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
One-sided noise spectral density	η_0	7.5×10^{-9}	V ² /GHz
Noise-equivalent quantization bits	N_{qb}	6	—
Quantization clipping probability	P_{qc}	10^{-7}	
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI

Table 179–24—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	6	—
Number of fixed-position taps	N_{fix}	15 ^a	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index	N_{max}	87 ^b	—
Normalized upper limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{max}(j)$	0.7 0.05	— —
Normalized lower limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{min}(j)$	−0.7 −0.05	— —
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	2×10^{-4}	—
Include maximum likelihood sequence detection		yes	—

^a There are 8 fixed-position post-cursor taps.

^b The latest post-cursor position for a floating tap is 80.

179.11.4.2 Channel signal and crosstalk path calculations

The channel paths between TP0d and TP5d used for calculation of the cable assembly COM consist of measured cable assembly signal and crosstalk paths, a reference transmitter model, and a reference receiver model. Reference transmitter and receiver models are defined in 178A.1.4 and 178A.1.5, respectively, using the parameters in Table 179–22 and Table 179–23, which define a model for each of the three host classes HL, HN, and HH.

179.11.4.2.1 Channel signal path

For a valid combination of a cable assembly class, transmitter host class, and receiver host class (see Table 179–21), the S-parameters of the channel signal path from TP0d to TP5d are calculated using Equation (178A–2), where

- S is the measured cable assembly signal differential-mode scattering matrix (TP1 to TP4).
- The transmitter and receiver S-parameter models $S^{(t)}$ and $S^{(r)}$ are calculated using the method described in 178A.1.4 and 178A.1.5, respectively, using the parameters in Table 179–22 and Table 179–23.
- The parameter $z_p^{(h)}$ for the transmitter is taken from the signal path column and the host class of the transmitter in Table 179–23.
- The parameter $z_p^{(h)}$ for the receiver is taken from the signal path column and the host class of the receiver in Table 179–23.

179.11.4.2.2 Channel crosstalk paths

The MDI is a significant contributor to crosstalk and is included in and characterized by the cable assembly crosstalk measurements. Crosstalk includes a near-end path where the aggressor is the PMD transmitter, and in some cases, additional near-end and far-end crosstalk paths where the aggressors are other PMD transmitters that are connected to the same cable assembly.

For a valid combination of a cable assembly class, transmitter host class, and receiver host class (see Table 179–21), the S-parameters of each channel near-end crosstalk path from TP0d to TP5d are calculated using Equation (178A–2), as specified in 179.11.4.2.1, with the following exceptions:

- S is the measured cable assembly near-end crosstalk differential-mode scattering matrix (TP1 to TP4).
- The parameter $z_p^{(h)}$ for the transmitter is taken from the signal path column and the host class of the receiver, instead of the transmitter, in Table 179–23.

For a valid combination of a cable assembly class, transmitter host class, and receiver host class (see Table 179–21), the S-parameters of each channel far-end crosstalk path from TP0d to TP5d are calculated using Equation (178A–2), as specified in 179.11.4.2.1, with the exception that S is the measured cable assembly far-end crosstalk differential-mode scattering matrix (TP1 to TP4).

Cable assemblies have several types, as described in Annex 179D. The number of near-end and far-end crosstalk paths per cable assembly type is given in Table 179–25.

Table 179–25—Number of crosstalk paths used in calculation of COM

Victim (one end)	NEXT	FEXT (other end)	
		QSFP224	QSFP-DD1600 or OSFP1600
QSFP224	4	3	7
QSFP-DD1600 or OSFP1600	8	7	7

179.11.5 Channel signal to common-mode ratio

Channel signal to common-mode ratio, $SCMR_{CH}$, is calculated using the procedure in 178A.2 with $P_{\text{peak}} = 10^{-7}$ and other parameter values taken from Table 179–24.

179.12 MDI specifications

The MDI couples the PMD (specified in 179.8 and 179.9) to the cable assembly (specified in 179.11).

Annex 179C specifies the MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8.

- 200GBASE-CR1 has three specified MDI connectors: QSFP224, QSFP-DD1600, and OSFP1600.
- 400GBASE-CR2 has three specified MDI connectors: QSFP224, QSFP-DD1600, and OSFP1600.
- 800GBASE-CR4 has three specified MDI connectors: QSFP224, QSFP-DD1600, and OSFP1600.
- 1.6TBASE-CR8 has two specified MDI connectors: QSFP-DD1600 and OSFP1600.

NOTE—Implementations that support multiple combinations of PHY types on the same MDI connector, as described in Annex 179C, must be configured appropriately for interoperability with the connected link partners. Selecting the appropriate configuration requires knowledge of the link partners.

179.13 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 14.7.

179.14 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 179–26 and Table 179–27. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 179–26, Table 179–27, Table 178B–3, and Table 178B–4.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 179–26—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	179.8.10	1.0.15	45.2.1.1
Global_PMD_transmit_disable	179.8.6	1.9.0	45.2.1.8
PMD_transmit_disable_7, PMD_transmit_disable_6, PMD_transmit_disable_5, PMD_transmit_disable_4 ^a	179.8.7	1.9.8:5	45.2.1.8
PMD_transmit_disable_3, PMD_transmit_disable_2 ^b	179.8.7	1.9.4:3	45.2.1.8
PMD_transmit_disable_1 ^c	179.8.7	1.9.2	45.2.1.8
PMD_transmit_disable_0	179.8.7	1.9.1	45.2.1.8

^a Available only in 1.6TBASE-CR8.

^b Available only in 800GBASE-CR4 and 1.6TBASE-CR8.

^c Available only in 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8.

Table 179–27—PMD status variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	179.8.8	1.1.7	45.2.1.2
PMD_transmit_fault	179.8.8	1.8.11	45.2.1.7
PMD_receive_fault	179.8.8	1.8.10	45.2.1.7
PMD_signal_detect_7, PMD_signal_detect_6, PMD_signal_detect_5, PMD_signal_detect_4 ^a	179.8.5	1.10.8:5	45.2.1.9

Table 179–27—PMD status variables and MDIO mapping (*continued*)

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_signal_detect_3, PMD_signal_detect_2 ^b	179.8.5	1.10.4:3	45.2.1.9
PMD_signal_detect_1 ^c	179.8.5	1.10.2	45.2.1.9
PMD_signal_detect_0	179.8.5	1.10.1	45.2.1.9
Global_PMD_signal_detect	179.8.4	1.10.0	45.2.1.9

^a Available only in 1.6TBASE-CR8.
^b Available only in 800GBASE-CR4 and 1.6TBASE-CR8.
^c Available only in 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8.

179.15 Protocol implementation conformance statement (PICS) proforma for Clause 179, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8⁸

179.15.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 179, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

179.15.2 Identification

179.15.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

179.15.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 179, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁸Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

179.15.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*PMD	PMD or PHY	179.1	Item marked with PMD are PMD or PHY specifications, these items are not applicable to cable assemblies.	O/1	Yes ☐ No ☐
*CBL	Cable assembly	179.11	Item marked with CBL are cable assembly specifications, these items are not applicable to PMDs or PHYs.	O/1	Yes ☐ No ☐
CR1	200GBASE-CR1	179.1	Can operate as a 200GBASE-CR1 PMD or PHY	PMD:O.2	Yes ☐ No ☐ N/A ☐
CR2	400GBASE-CR2	179.1	Can operate as a 400GBASE-CR2 PMD or PHY	PMD:O.2	Yes ☐ No ☐ N/A ☐
CR4	800GBASE-CR4	179.1	Can operate as an 800GBASE-CR4 PMD or PHY	PMD:O.2	Yes ☐ No ☐ N/A ☐
CR8	1.6TBASE-CR8	179.1	Can operate as a 1.6TBASE-CR8 PMD or PHY	PMD:O.2	Yes ☐ No ☐ N/A ☐
HL	Host class HL	179.9	PMD conforms to host class HL transmitter and receiver requirements on all MDI lanes	PMD:O.3	Yes ☐ No ☐ N/A ☐
HN	Host class HN	179.9	PMD conforms to host class HN transmitter and receiver requirements on all MDI lanes	PMD:O.3	Yes ☐ No ☐ N/A ☐
HH	Host class HH	179.9	PMD conforms to host class HH transmitter and receiver requirements on all MDI lanes	PMD:O.3	Yes ☐ No ☐ N/A ☐
PHY	Single package PHY	179.1	PMD and PCS are in the same package	PMD:O/4	Yes ☐ No ☐ N/A ☐
AUIC2C	AUI-n C2C	179.1	PMD is separated from the PCS by an AUI-C2C specified in Annex 120B, Annex 120D, Annex 120F, or Annex 176C	PMD:O/4	Yes ☐ No ☐ N/A ☐
nGMII	200GMII, 400GMII, 800GMII, or 1.6TMII	179.1	Interface is implemented or functionally equivalent	PMD:O	Yes ☐ No ☐ N/A ☐
AN	Auto-Negotiation	179.1	Auto-Negotiation is implemented	PMD:M	Yes ☐ N/A ☐
PMDAPSU	APSU functions in PMD	179.8.9	APSU functions are implemented in the PMD	PMD:M	Yes ☐ N/A ☐

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO capability	179.14	Clause 45 MDIO is implemented with the management variable mapping in 179.14	PMD:O	Yes [] No [] N/A []
TS	Time Synchronization	90	Support for Time Synchronization is implemented	PMD:O	Yes [] No [] N/A []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

179.15.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179.15.4.1 PHY specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PS1	Service Interface	179.4	PMD service interface implemented	PMD:M	Yes [] N/A []
PS2	Delay constraints	179.6	Complies with the delay constraints	PMD:M	Yes [] N/A []
PS3	Skew constraints	179.7	Complies with the skew constraints	PMD:M	Yes [] N/A []
PS4	PMD functional specifications	179.8	Complies with all PMD functional specifications	PMD:M	Yes [] N/A []
PS5	PMD electrical specifications	179.9	Complies with all PMD electrical specifications	PMD:M	Yes [] N/A []

179.15.4.2 MDI specifications

Item	Feature	Subclause	Value/Comment	Status	Support
MDI1	QSFP224	179.12	PMD or cable assembly uses MDI connector type QSFP224	O/5	Yes [] No [] N/A []
MDI2	QSFP-DD1600	179.12	PMD or cable assembly uses MDI connector type QSFP-DD1600	O/5	Yes [] No [] N/A []
MDI3	OSFP-1600	179.12	PMD or cable assembly uses MDI connector type OSFP-1600	O/5	Yes [] No [] N/A []
MDI4	Mixed MDIs	179.12	Cable assembly uses different MDI connector types on each end	CBL: O/5	Yes [] No [] N/A []

179.15.4.3 Cable assembly specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CA1	Cable assembly type 200GBASE-CR1	179.11	Supports single-lane links between two 200GBASE-CR1 PHYs	CBL:O/6	Yes [] No [] N/A []
CA2	Cable assembly type 400GBASE-CR2	179.11	Supports two-lane links between two 400GBASE-CR2 PHYs	CBL:O/6	Yes [] No [] N/A []
CA3	Cable assembly type 800GBASE-CR4	179.11	Supports four-lane links between two 800GBASE-CR4 PHYs	CBL:O/6	Yes [] No [] N/A []
CA4	Cable assembly type 1.6TBASE-CR8	179.11	Supports eight-lane links between two 1.6TBASE-CR8 PHYs	CBL:O/6	Yes [] No [] N/A []
CA5	Cable assembly class CA-A	179.11	Meets the requirements of cable assembly class CA-A	CBL:O.7	Yes [] No [] N/A []
CA6	Cable assembly class CA-B	179.11	Meets the requirements of cable assembly class CA-B	CBL:O.7	Yes [] No [] N/A []
CA7	Cable assembly class CA-C	179.11	Meets the requirements of cable assembly class CA-C	CBL:O.7	Yes [] No [] N/A []
CA5	Cable assembly class CA-D	179.11	Meets the requirements of cable assembly class CA-D	CBL:O.7	Yes [] No [] N/A []
CA6	AC-coupling	179.11	All lanes are AC-coupled within the cable assembly	CBL:M	Yes [] N/A []

179.15.4.4 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Environmental specifications	179.13	Conform to applicable requirements of 14.7	M	Yes []

179.15.4.5 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	179.14		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	179.14		MD:M	Yes [] N/A []